When Do Robot Policies Learn to Listen?

A Controlled, Pre-Registered Study of Training-Data Structure and Language Grounding

nonberw
github.com/nonberw/listen

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Abstract

Language-conditioned robot policies often ignore their instructions and execute whatever task they learned for the scene in front of them. The usual explanation is the structure of robot datasets, in which a scene co-occurs with only one or a few tasks. This explanation has not been tested causally. We train visuomotor imitation policies from scratch in a controlled simulated pick-up task. Total data (1200 demonstrations), expert, architecture and training recipe are held fixed, and the only variable is the fraction f of *contrastive* scenes, demonstrated with several instructions. We then ask the policies for blocks that a scene was never demonstrated with. Following a pre-registered protocol, we find four things. (i) Contrastive data in some scenes transfers grounding to others. The counterfactual grounding rate rises monotonically from 48.9% at f=0 to 61.3% at f=0.25 (p=0.033, Holm) and 71.3% at f=0.5 ($p<10^{-4}$), with no loss in nominal success. (ii) Small doses, two to four contrastive scenes, have no detectable effect. (iii) Contrary to our pre-registered hypothesis, failures at f=0 are rarely shortcut executions (1.4%). They are hesitations: the gripper descends about 5 cm from the instructed block, displaced toward the scene's trained block. (iv) In an exploratory experiment with only four scenes, where memorizing scenes is easy, the familiar VLA failure appears. Grounding collapses to 0%, the shortcut rate rises to 17%, and one contrastive scene out of four does not help. Language blindness therefore depends on how easy the scene shortcut is to learn relative to the language, and remedies need contrastive data spread across many scenes.

1 Introduction

Language-conditioned robot policies promise that a single model can be told what to do. A growing body of evidence suggests they often do not listen. When the instruction is changed but the scene is not, vision-language-action (VLA) models frequently execute the task they were trained on in that scene, and sometimes perform almost as well with an empty instruction [1, 2, 7, 18]. The standard explanation is about data: in robot datasets a scene co-occurs with only one or a few tasks, so the image alone predicts the action and the instruction carries little additional information [1, 7].

This explanation is plausible, but it has not been tested causally. Existing studies evaluate or probe large pretrained models whose training data cannot be changed, and they are explicit about it. Jiang et al. [7] report their findings "without establishing a causal mechanism" and list training that forces instructions to choose between competing behaviors in the same scene as future work. Mechanistic analyses [7, 15] show *where* instruction information is lost inside a model, but not *why* training puts it there.

We test the data explanation directly. In a controlled simulated pick-up task we train visuomotor policies from scratch on datasets that differ in exactly one property: the fraction f

of *contrastive* scenes, meaning scenes demonstrated with several different instructions. The total amount of data, the expert, the architecture and the training recipe are held fixed. We then measure whether the policies follow *counterfactual* instructions: requests for a block that the scene was never demonstrated with. All hypotheses, metrics and the analysis plan were pre-registered before any experiment was run, and every deviation is documented.

Contributions.

- A controlled, pre-registered test of the data-structure explanation for instruction blindness, with a dose–response curve over the contrastive fraction at a fixed data budget.
- Evidence that grounding *transfers*: contrastive demonstrations in some scenes raise instruction following in scenes that were only ever shown with one task.
- A failure analysis showing that the dominant failure mode can be *hesitation* rather than shortcut execution, and when it turns into the shortcut. An exploratory few-scene experiment recovers the VLA failure mode, which suggests that the relative learnability of the scene and the language cue is the deciding factor.
- Open code, the pre-registration with its commit history, and per-episode records of all evaluation episodes.

2 Related work

Instruction blindness in VLAs. LIBERO-Plus [2] and LIBERO-PRO [18] perturb LIBERO tasks [10] and find that leading models are largely insensitive to the instruction. LIBERO-CF [1] builds counterfactual instructions for training scenes, names the failure "vision overriding language", and proposes an inference-time guidance fix. Jiang et al. [7] show that the instructed target is often linearly decodable from a policy's internal activations even when the policy does not act on it, and Wulff and Cangelosi [15] localize language sensitivity with activation patching. Zhang and Bisk [16] study how fine-tuning erodes instruction following. All of these evaluate existing models. We complement them with a controlled training experiment.

Language priors and contrastive data outside robotics. The same failure appeared in visual question answering, where models answered from the question alone. The fix was to balance the dataset with complementary pairs: the same question with different images and answers [5]. Contrast sets [3] and analyses of annotation artifacts [6] made the same point for text. These are all instances of shortcut learning [4]. Our contrastive scenes are the embodied analogue of complementary pairs, with the roles of image and text swapped: the same image with different instructions and actions.

Imitation learning machinery. We use action chunking with temporal ensembling [17], FiLM conditioning [11], a frozen sentence encoder [12, 14], DART noise injection [9] and random-shift augmentation [8], in MuJoCo [13].

3 Experimental design

Task. A 6-DoF arm with a two-finger gripper faces three colored blocks on a table, with colors drawn from a pool of four. The instruction asks for one block by color, in one of five training

paraphrases (e.g. "grab the red block"). The policy sees a 64×64 RGB image and its end-effector position and gripper width. At 10 Hz it outputs end-effector displacements and an open/close command, which a low-level IK controller tracks.

Scenes and contrastive fraction. There are 40 scene templates, each a fixed set of three colors at fixed nominal positions, with a ±2 cm and ±15° jitter per episode. A *non-contrastive* scene is always demonstrated with the same target. A *contrastive* scene is demonstrated with each of its three blocks equally often. The default targets of non-contrastive scenes are balanced over colors (9 to 11 scenes per color), so a global color prior cannot explain behavior. The only cues left are the scene's identity and the instruction. We vary $f \in \{0, 0.05, 0.1, 0.25, 0.5, 1\}$. Contrastive sets are nested across f, and scenes 20–39 are non-contrastive for every $f \leq 0.5$. These scenes serve as the evaluation set. Every condition uses exactly $N = 1200$ demonstrations (30 per scene), sampled from a shared pool of scripted expert demonstrations collected with DART noise ($\sigma = 4$ mm).

Policy and training. A four-layer CNN in which every layer is FiLM-modulated by the instruction embedding (frozen `all-MiniLM-L6-v2`, 384-d), followed by an MLP head that predicts a chunk of five future actions. Training uses 12k AdamW steps with batch 256, MSE loss and random-shift augmentation, and is identical across conditions. Execution uses temporal ensembling over the last five chunks.

Metrics. The primary metric is the *counterfactual grounding rate* (CGR). In a non-contrastive evaluation scene, the policy is asked for one of the two blocks the scene was never demonstrated with, and CGR is the fraction of episodes in which the first block the gripper closes on is the instructed one. We also report the *default rate* (closing on the scene's trained block), the rate of committing to no block, nominal success on the trained instruction, CGR with three held-out paraphrases, and grounding on ten never-seen scene layouts. Each run is evaluated on the same 660 episodes (300 counterfactual, 150 nominal, 120 paraphrase, 90 held-out-layout).

Statistics. There are three training seeds per condition. Confidence intervals come from a two-level bootstrap (training runs, then episodes). Each $f > 0$ is compared with $f = 0$ by a one-sided run-level bootstrap test with Holm correction, together with a logistic regression of grounding on categorical f with a fixed seed effect, as pre-registered.

4 Results

4.1 Pilot

The pilot (seed 99, $f \in \{0, 1\}$, excluded from all tests) exposed a manipulation failure that had nothing to do with language. When only the first action of each chunk was executed, policies reached the block with 2 mm accuracy but almost never closed the gripper (1–3% nominal success). The states just before and just after the expert closes look the same, and the regressed gripper command settled around −0.5. Temporal ensembling [17] fixed this (80–87% nominal success on a 30-episode check). We adopted it as a pre-declared pilot-stage deviation before the main sweep. The pilot also suggested that our first hypothesis would fail, which motivated experiment E (Section 4.4). That experiment was registered before any main-sweep result was seen.

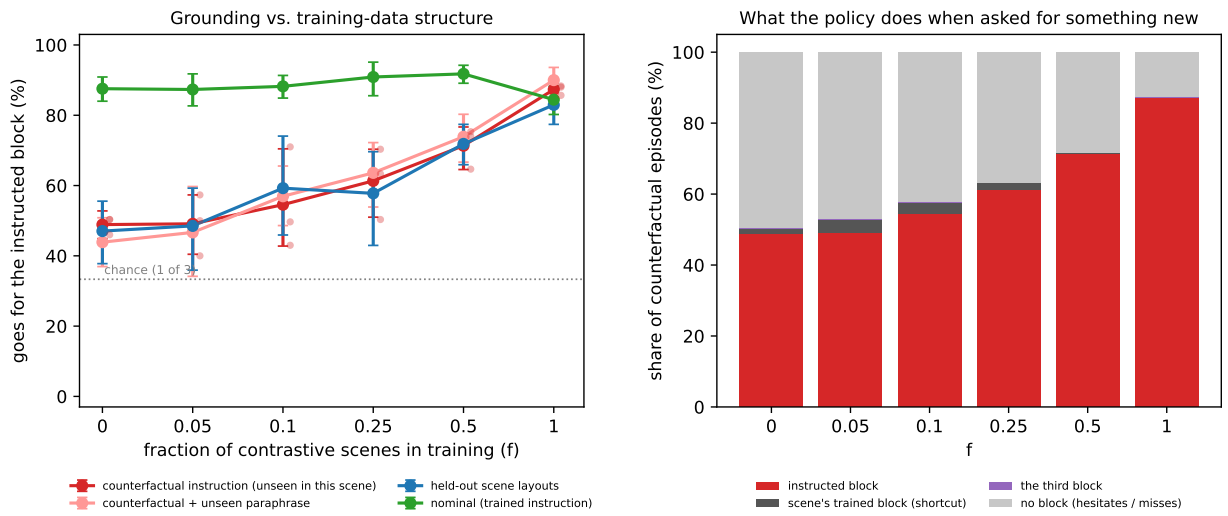


Figure 1: **Left:** rate of going for the instructed block against the contrastive fraction f. The counterfactual curves (red) are measured in scenes that are non-contrastive for every $f \leq 0.5$, so for $f \leq 0.5$ any gain is transfer from *other* scenes. Error bars are 95% two-level bootstrap CIs and small dots are individual training runs. At f=1 these scenes are themselves contrastive, so that point is an in-distribution reference, not transfer. **Right:** outcome of counterfactual episodes. Taking the scene's trained block (the "shortcut") is rare at every f, and failures are almost all "no block".

4.2 Contrastive data transfers grounding

Figure 1 and Table 1 show the main result. Counterfactual grounding in the non-contrastive evaluation scenes increases monotonically with f, from 48.9% at f=0 to 71.3% at f=0.5. These twenty scenes were demonstrated with a single target in every one of these conditions, so the gain comes entirely from contrastive data *elsewhere*. The same trend appears with paraphrases never seen in training and on scene layouts never seen at all.

Table 2 gives the pre-registered tests. The effect is significant at f=0.25 (10 contrastive scenes) and f=0.5 (20 scenes), but not at f=0.05 or f=0.1 (2 and 4 scenes). The pre-registered logistic regression with a fixed seed effect agrees in direction, with log-odds coefficients of 0.01, 0.23, 0.51 and 0.97 for $f = 0.05 \ldots 0.5$, and it calls f=0.1 significant (p=0.015). That model treats episodes as independent and ignores within-run correlation, so we consider the run-level test the more trustworthy one, and we do not claim an effect at f=0.1. Seed-to-seed variation is large: at f=0.1 the three runs reach 43%, 50% and 71%. No $f < 1$ reaches 80% counterfactual grounding, so the pre-registered f_{80} is 1.

Hypotheses. **H1** (low grounding and frequent shortcut at f=0) is *rejected*: grounding is 48.9% [44.9, 52.8], well above chance, and the shortcut rate is 1.4%. **H2** (monotonic increase) is supported. **H3** (transfer at $f \in \{0.1, 0.25\}$) is supported at f=0.25 and not at f=0.1. **H4** (nominal success drops by at most 10 points) is supported: the largest drop is 3.2 points, at f=1.

	counterfactual (non-contrastive scenes)			paraphrase	held-out	nominal
f	grounded [95% CI]	shortcut	none	grounded	grounded	success [95% CI]
0	48.9 [44.9, 52.8]	1.4	49.3	43.9	47.0	87.6 [84.0, 90.9]
0.05	49.1 [40.4, 57.3]	3.6	46.9	46.7	48.5	87.3 [82.7, 91.8]
0.1	54.6 [42.8, 70.4]	3.0	42.1	56.9	59.3	88.2 [84.9, 91.3]
0.25	61.3 [51.0, 70.3]	1.9	36.8	63.6	57.8	90.9 [85.6, 95.1]
0.5	71.3 [64.6, 76.7]	0.4	28.2	73.9	71.9	91.8 [89.1, 94.2]
1	87.3 [84.8, 89.8]	0.0	12.7	90.0	83.0	84.4 [80.2, 88.2]

Table 1: Main results, in %. There are 3 training runs per row; each run is evaluated on 300 counterfactual, 120 paraphrase, 90 held-out-layout and 150 nominal episodes.

f vs. 0	Δ grounding (pts)	p (one-sided)	p (Holm)
0.05	+0.2	0.4549	0.5346
0.1	+5.7	0.2673	0.5346
0.25	+12.4	0.0111	0.0333
0.5	+22.4	0.0000	0.0000

Table 2: Pre-registered run-level bootstrap tests (10^4 resamples; 0.0000 means no resample reached $\Delta \leq 0$).

4.3 How policies fail: hesitation, not stubbornness

Across all conditions, when a policy commits to a block it is almost always the instructed one. The third block, never demonstrated in the scene and not named, is chosen in at most 0.4% of counterfactual episodes. The failures are "no block" episodes. To see what happens in them, we replayed the first 120 counterfactual episodes of one f=0 run and one f=0.5 run and logged the gripper path. At f=0, 63 of 120 episodes committed to nothing. In those episodes the gripper still descended to the table (median lowest height 1.4 cm), but it ended up a median 4.9 cm from the instructed block and 10.5 cm from the trained block, which on average sit 14.2 cm apart. In 71% of these episodes it was closer to the instructed block. At f=0.5, 45 of 120 episodes committed to nothing, and the miss shrank to 2.9 cm from the instructed block. The policy reads the instruction well enough to head the right way, but it is pulled partway toward the scene's trained block and misses the 3 cm commit radius. This is consistent with a mean-squared-error policy averaging two attractors, one from the instruction and one from scene memory. It is also the behavioral counterpart of the "encoded but not in control" finding of Jiang et al. [7]. Contrastive data reduces the pull.

4.4 Exploratory: with few scenes, the shortcut comes back

Why did f=0 yield substantial grounding instead of the shortcut that VLAs show? In our main design, default targets are balanced over colors across 40 scenes, so the scene shortcut means memorizing 40 scene-to-target pairs, while the language shortcut is a single rule: match the color word to the color. We expected the shortcut to return when scene memorization is easy, and tested this with four scenes and the same 1200 demonstrations (Table 3, Figure 2). This experiment is exploratory: it was added after the pilot, but registered before the main results were seen.

With four scenes, counterfactual grounding collapses to zero (0 of 900 episodes), and the shortcut rate rises from 1.4% to 17.0% (E1; run-level bootstrap, $p<10^{-4}$). Nominal success *increases* to

	counterfactual				nominal
Regime	grounded	shortcut [95% CI]	third block	none	success
40 scenes, $f = 0$	48.9	1.4 [0.3, 2.7]	0.3	49.3	87.6
4 scenes, $f = 0$	0.0	17.0 [10.1, 24.2]	8.7	74.3	97.8
4 scenes, $f = 0.25$	0.1	29.0 [22.2, 38.0]	0.1	70.8	98.9

Table 3: Experiment E (exploratory), in %. There are 3 runs per row. Counterfactual episodes are in the three scenes that are non-contrastive in both few-scene conditions.

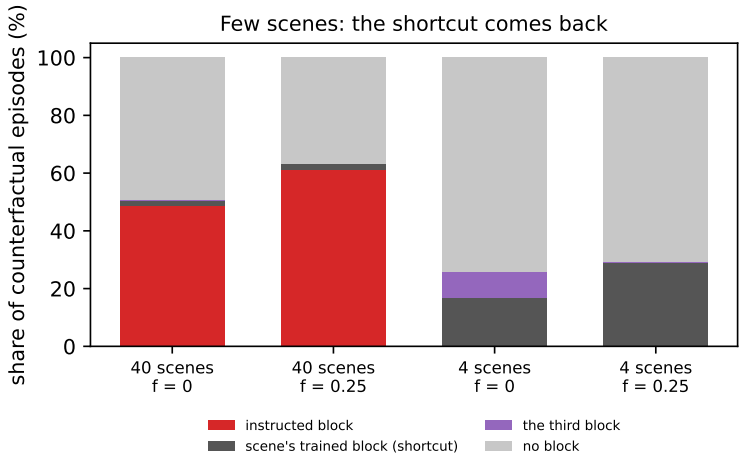


Figure 2: Counterfactual outcomes with 40 and with 4 scenes. With four scenes the policy never follows a counterfactual instruction, and the shortcut reappears.

97.8%: the policies have memorized the scenes very well. Our second expectation (E2) failed. Making one of the four scenes contrastive did not reduce the shortcut in the other three; if anything it rose to 29.0% (p=0.99 for a reduction). One contrastive scene can be solved by memorizing that scene too (scene plus word gives the target) without learning the general rule. This matches the main sweep, where two or four contrastive scenes had no detectable effect and ten did.

4.5 The frozen language encoder is itself partly color-blind

In the frozen MiniLM embedding space, "pick up the red block" is closer to "pick up the blue block" than to "grab the red cube". Across our 8 templates and 4 colors, the mean cosine similarity is 0.80 for same template and different color, versus 0.66 for same color and different template. In a variance decomposition, the template explains 66% of the embedding variance and the color only 32%. The policy must learn to amplify a minority direction of its own input, which is one more reason the scene can win. Fine-tuning or replacing the encoder is a natural follow-up we did not run.

5 Discussion

What decides whether a policy listens? Our results support the data-structure explanation [1, 7], and they also refine it. "One task per scene" does not cause language blindness by itself.

In our 40-scene world, every scene had one task at f=0, and the policies still followed counterfactual instructions about half the time. What decided the outcome was whether the scene cue was *easier to learn* than the language cue. With many scenes and a simple, shared language rule, the language cue wins. With few scenes, the scene cue wins completely. Robot benchmarks such as LIBERO [10] have few scenes and rich, task-specific instructions, which puts them in the second regime. This view is the standard shortcut-learning account [4] applied to the competition between two input modalities.

How much contrastive data, and where? The practical message is that contrastive data works and transfers. A modest share of scenes demonstrated with several instructions improves instruction following everywhere, at no cost to nominal performance. But it has to be spread over *many* scenes: in both experiments, one to four contrastive scenes did nothing detectable. Because f and the number of contrastive scenes are perfectly correlated in our main sweep, we cannot separate the two. Experiment E points to the number of distinct contrastive scenes as the factor that matters. This is the direct analogue of complementary pairs in VQA [5], which worked because they covered the whole dataset.

Hesitation as a measurable failure mode. Counterfactual evaluations usually score success or "did the original task". Our policies mostly did neither. Measuring where the gripper ends up separates a policy that ignores language from one that reads it but cannot commit, and the two call for different remedies: more contrastive data or a multimodal action head, respectively.

6 Limitations

- **Scale.** Small policies trained from scratch in simulation, not pretrained VLAs. The causal question needs control over the training data, which only this scale allows on our budget. Whether the dose–response transfers to large pretrained models is untested.
- **Language.** Instructions differ only in a color word and a paraphrase template. Real instructions vary in verbs, objects and spatial relations, and the language cue is then harder to learn, which by our own account should make the shortcut stronger.
- **Fraction vs. number of scenes.** These are confounded in the main sweep. The few-scene experiment suggests that the number of scenes matters, but it is exploratory and has only two conditions.
- **Statistics.** Three runs per condition, with large between-run variance. The episode-level regression and the run-level test disagree at f=0.1, and we report the conservative one.
- **Execution.** Temporal ensembling was adopted after the pilot (documented). The MSE action head may make hesitation more likely than a multimodal head would.
- **Single task family**, one robot, one camera, perfect simulated rendering.

7 Conclusion

We tested causally the common explanation for why language-conditioned robot policies ignore their instructions. Holding the data budget fixed, contrastive demonstrations, meaning the same scene with different instructions, raised instruction following even in scenes that never had any,

with no loss of skill. Small doses had no detectable effect, and a single contrastive scene was not enough. Policies did not simply execute the trained task: they hesitated between the instruction and the scene's memory. With few scenes, the familiar blindness returned in full. For anyone collecting robot data, the recipe is to show the same scenes with different instructions, and to do it across many scenes.

Reproducibility. Code, the pre-registration with its commit history, all per-episode evaluation records and the analysis script are available at `https://github.com/nonberw/listen`.

References

[1] Yu Fang, Yuchun Feng, Dong Jing, Jiaqi Liu, Yue Yang, Zhenyu Wei, Daniel Szafir, and Mingyu Ding. When vision overrides language: Evaluating and mitigating counterfactual failures in VLAs. *arXiv preprint arXiv:2602.17659*, 2026.

[2] Senyu Fei, Siyin Wang, Junhao Shi, Zihao Dai, Jikun Cai, Pengfang Qian, Li Ji, Xinzhe He, Shiduo Zhang, Zhaoye Fei, Jinlan Fu, Jingjing Gong, and Xipeng Qiu. LIBERO-Plus: In-depth robustness analysis of vision-language-action models. *arXiv preprint arXiv:2510.13626*, 2025.

[3] Matt Gardner, Yoav Artzi, Victoria Basmov, Jonathan Berant, Ben Bogin, et al. Evaluating models' local decision boundaries via contrast sets. In *Findings of the Association for Computational Linguistics: EMNLP*, 2020.

[4] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.

[5] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2017.

[6] Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel R. Bowman, and Noah A. Smith. Annotation artifacts in natural language inference data. In *Conference of the North American Chapter of the Association for Computational Linguistics*, 2018.

[7] Shaohan Jiang, Jiahang Cao, Qiduo He, Fengting Deng, Kun Wu, Jingkai Sun, Jiaxu Wang, Qiang Zhang, Qihao Zheng, Chunfeng Song, Ping Luo, and Andrew F. Luo. Encoded but not in control: Revealing the grounding gap in vision-language robot policies. *arXiv preprint arXiv:2610.06235*, 2026.

[8] Ilya Kostrikov, Denis Yarats, and Rob Fergus. Image augmentation is all you need: Regularizing deep reinforcement learning from pixels. In *International Conference on Learning Representations*, 2021.

[9] Michael Laskey, Jonathan Lee, Roy Fox, Anca Dragan, and Ken Goldberg. DART: Noise injection for robust imitation learning. In *Conference on Robot Learning*, 2017.

[10] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems*, 2023.

[11] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *AAAI Conference on Artificial Intelligence*, 2018.

[12] Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Conference on Empirical Methods in Natural Language Processing*, 2019.

[13] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012.

[14] Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, 2020.

[15] Theodor Wulff and Angelo Cangelosi. Do vision-language-action models understand instructions? A mechanistic interpretability study on language grounding. *arXiv preprint arXiv:2610.10178*, 2026.

[16] Haochen Zhang and Yonatan Bisk. Flatness preserves instruction following in vision-language-action models. *arXiv preprint arXiv:2606.23641*, 2026.

[17] Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Robotics: Science and Systems*, 2023.

[18] Xueyang Zhou, Yangming Xu, Guiyao Tie, Yongchao Chen, Guowen Zhang, Duanfeng Chu, Pan Zhou, and Lichao Sun. LIBERO-PRO: Towards robust and fair evaluation of vision-language-action models beyond memorization. *arXiv preprint arXiv:2510.03827*, 2025.